Course Code:	24CS305	Course Title:	Database Management System
Credits:	4	L – T – P	3-0-2

Course Objectives (COBs)

By the end of this course, students will be able to:

1. **Understand** the fundamental concepts of database systems, data models, and architectures.
2. **Apply** relational modeling concepts to design structured and normalized databases for real-world applications.
3. **Develop** efficient SQL queries to store, manipulate, and retrieve data from relational databases.
4. **Analyze** transaction management, concurrency control, and recovery techniques to ensure data consistency and reliability.
5. **Explore** indexing, query optimization, and NoSQL systems to build scalable and performance-oriented applications.

Teaching-Learning Process:

Suggested strategies that teachers may use to effectively achieve the course outcomes:

1. Chalk and Talk
2. Peer Learning
3. Interactive session
4. Hands-on Labs
5. Project based Learning

UNIT I – INTRODUCTION TO DATABASES AND DATA MODELING	[9 hours]
Purpose and need for databases, Limitations of file-based systems, Components of a DBMS , Types of database users and database system architecture, Data models: Hierarchical, Network, Relational, Object-oriented, Entity-Relationship (ER) Model: Entities, attributes, relationships ,Keys: primary, candidate, super, foreign keys, Extended ER concepts: generalization, specialization, aggregation ER to relational schema mapping, Case Study: Modeling a hospital or student information system	

UNIT II – RELATIONAL MODEL AND SQL	[9 hours]
Relational data model: schema, tuples, domains, integrity rules, Relational algebra: selection, projection, joins, set operations, Structured Query Language (SQL):DDL (CREATE, ALTER, DROP),DML (INSERT, UPDATE, DELETE),DCL, TCL (COMMIT, ROLLBACK, GRANT) Constraints: NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK, GROUP BY, HAVING, nested queries and views, Joins: inner, outer (left/right/full), self-joins, Indexes,	

sequences, and triggers (introductory).

UNIT III – RELATIONAL DATABASE DESIGN AND NORMALIZATION	[9 hours]
ER and EER-to-Relational mapping ,Update anomalies , Functional dependencies , Inference rules – Minimal cover – Properties of relational decomposition – Normalization – 1NF, 2NF, 3NF, BCNF, 4NF, 5NF , Case Study: Normalizing a retail inventory or library database	

UNIT IV – TRANSACTION	[9 hours]
Transaction concepts – ACID properties – Schedules – Serializability – Concurrency Control – Two-phase locking techniques – Deadlock – Occurrence and Recovery	

UNIT V – STORAGE, INDEXING, QUERY PROCESSING, AND EMERGING TRENDS	[9 hours]
Storage and file organization: heap, sorted, hashed files, Indexing: single-level, multi-level, clustered, B+, hash indexing, Query processing: parsing, optimization, and execution plans, Basics of query optimization and cost-based strategies **Distributed databases** – horizontal/vertical fragmentation, replication , Introduction to **NoSQL**: key-value stores, document stores, graph DBs, Real-world DBMS applications: e-commerce, banking, IoT, healthcare , Cloud-based databases: introduction to DBaaS (e.g., Firebase, Amazon RDS)	

Laboratory Component: **[30 hours]**

Any 12 experiments have to be completed from the following list of experiments.

S.No	**Name of the Exercises**
1	Design an ER model for a college library or hospital.
2	Create tables with primary key, foreign key, and check constraints.
3	To perform data insertion, deletion, and retrieval using basic SQL DML and SELECT queries.
4	To use aggregate functions and group data for analysis and reporting.
5	To retrieve data from multiple tables using different types of joins and subqueries.
6	Perform INNER JOIN, LEFT JOIN, and nested queries.
7	Normalize tables to eliminate redundancy.
8	To create views for data abstraction and indexes for performance improvement.
9	Demonstrate transaction behavior.
10	To simulate transactions and demonstrate concurrency control using SQL.
11	Compare query performance with and without indexes.
12	To explore basic operations in a NoSQL document database.

Course outcomes:

On completion of the course, the student will have the ability to:

CO No.	Course Outcomes	Cognitive Level
CO1	Apply data modeling concepts to design Entity-Relationship diagrams for real-world problems.	K3
CO2	Use SQL to create, query, and manage relational databases for practical applications.	K3
CO3	Apply normalization techniques to design efficient and consistent database schemas.	K3
CO4	Apply transaction control and concurrency techniques to ensure data consistency in multi-user environments..	K3
CO5	Apply indexing, query optimization, and NoSQL techniques to develop scalable and efficient database applications.	K3

COs and POs Mapping:

COs	Pos											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
CO1	3	2	2		1						1	
CO2	3	3	1	1	1							2
CO3	3	2	1	1	1							
CO4	3	2	1	1	1							
CO5	3	2	1	1	1							2

Level 3- Highly Mapped, Level 2- Moderately Mapped, Level 1- Low Mapped, Level 0- Not Mapped

Scheme of Evaluation:

<table>
<tr><th>Component</th><th>Type of assessment</th><th>Max Marks</th><th>Reduced Marks</th><th>Total</th><th>Final marks</th></tr>
<tr><td rowspan="5">Continuous Internal Examination (CIE) - Theory</td><td>CIE – I</td><td>100</td><td rowspan="2">50</td><td rowspan="5">100</td><td rowspan="5">25</td></tr>
<tr><td>CIE – II</td><td>100</td></tr>
<tr><td>MCQ</td><td>20</td><td>10</td></tr>
<tr><td>Skill Assessment - I</td><td>40</td><td rowspan="2">40</td></tr>
<tr><td>Skill Assessment - II</td><td>40</td></tr>
<tr><td rowspan="2">Continuous Internal Examination (CIE) - Laboratory</td><td>Continuous Assessment</td><td>75</td><td>75</td><td rowspan="2">100</td><td rowspan="2">25</td></tr>
<tr><td>Model Lab Exam</td><td>25</td><td>25</td></tr>
<tr><td>End Semester</td><td>Theory Exam</td><td>100</td><td>35</td><td>50</td><td>50</td></tr>
</table>

Examination (ESE)	Lab Exam	100	15		
				Total	100

Assessment Pattern:

Bloom's Category	Continuous Assessment Tests		Terminal Examination
	1	2	
Remember	20	20	20
Understand	40	40	40
Apply	40	40	40
Analyze	0	0	0
Evaluate	0	0	0
Create	0	0	0

End semester Examination: (QP PATTERN)

- Each unit consists of two 2 marks questions and one 16 marks question (either or).
- All the fifteen questions have to be answered.

Text Books:

1. Abraham Silberschatz, Henry F Korth, S Sudharshan, "Database System Concepts", 6th Edition, Tata Mc Graw Hill, 2011.
2. 2. Ramez Elmasri, Shamkant B. Navathe, Fundamentals of Database Systems, 7th Edition, Pearson, 2017.

Reference Books:

1. Toby Teorey, Sam Lightstone, Tom Nadeau, H. V. Jagadish, "Database Modeling And Design – Logical Design", Fifth Edition, Morgan Kaufmann Publishers, 2011.
2. Carlos Coronel, Steven Morris, and Peter Rob, Database Systems: Design, Implementation, and Management, Ninth Edition, Cengage learning, 2012
3. Thomas M. Connolly, Carolyn E. Begg, Database Systems – A Practical Approach to Design, Implementation, and Management, Sixth Edition, Global Edition, Pearson Education, 2015.
4. Hector Garcia-Molina, Jeffrey D Ullman, Jennifer Widom, "Database Systems:The Complete Book", 2nd edition, Pearson.

Web Links and Video Lectures (E-Resources):

1. ER Model: https://byjus.com/gate/entity-relationship-model-in-dbms-notes/
2. SQL: https://www.w3schools.com/sql/
3. Normalization: https://www.geeksforgeeks.org/introduction-of-database-normalization/
4. ACID Properties: https://www.geeksforgeeks.org/introduction-of-database-normalization/

5. Mongo DB: https://www.mongodb.com/

Activity-Based Learning /Practical-Based Learning:

1. http://nptel.ac.in
2. https://swayam.gov.in

Suggested Skill Activities:

1. A company has 10 employees. Create a table to store their personal details and salary details.
2. The college comprises 6 departments. Write SQL query to find the topper in each semester in each department.
3. The college comprises 6 departments. Use joins to help transport office to display the name of students from same place.
4. Write PL/SQL program to display the result of students and include trigger a message when the result is pass.
5. Write SQL query to display the toppers in the current semester and order by department.
6. To design and implement a complete database system for a selected real-world application.
7. Design and implement a database for a small system (e.g., Inventory, Student Portal).
8. Create an ER diagram using online tools (e.g., dbdiagram.io, Lucidchart).
9. Write complex SQL queries for a sample business database (e.g., Inventory or HR system).
10. Normalize a messy spreadsheet into 3NF using Google Sheets/Excel and convert it into SQL schema.
11. Load and query data in MongoDB or Firebase (e.g., student records, to-do list).

Course Code:	24ML351	Course Title:	Mini Project – I (Fundamentals of AI & Machine Learning through Design and Simulation)
Credits:	1	L – T – P	0-0-2

Course objectives

- To enable students to apply foundational programming and data science principles to real-world problems.
- To enhance logical reasoning, problem-solving, and technical implementation skills.
- To simulate simple AI-based systems through design and experimentation.
- To prepare students for future AI projects by nurturing a design-first mindset.

Problem Identification & Analysis:`	(5 hours)
• Select a problem that is relevant and feasible within the scope of AI/ML. • Ensure the problem has sufficient data available for analysis.	
Solution Design:	**(10 hours)**
• Define the objectives and scope of the project. • Choose appropriate AI/ML algorithms and tools. • Design the system architecture and workflow.	
Implementation and Simulation:	**(13 hours)**
• Develop the solution using programming languages like Python. • Utilize libraries such as TensorFlow, scikit-learn, or Keras. • Integrate data preprocessing, model training, and evaluation components • Assess the performance using metrics like accuracy, precision, recall, or F1-score. • Compare results with baseline or existing methods.	
Documentation	**(2 hours)**
• Prepare a detailed report (20-25 pages) covering: Introduction and problem statement. Literature review. Methodology and design. Implementation details. Results and discussion. Conclusion and future work.	